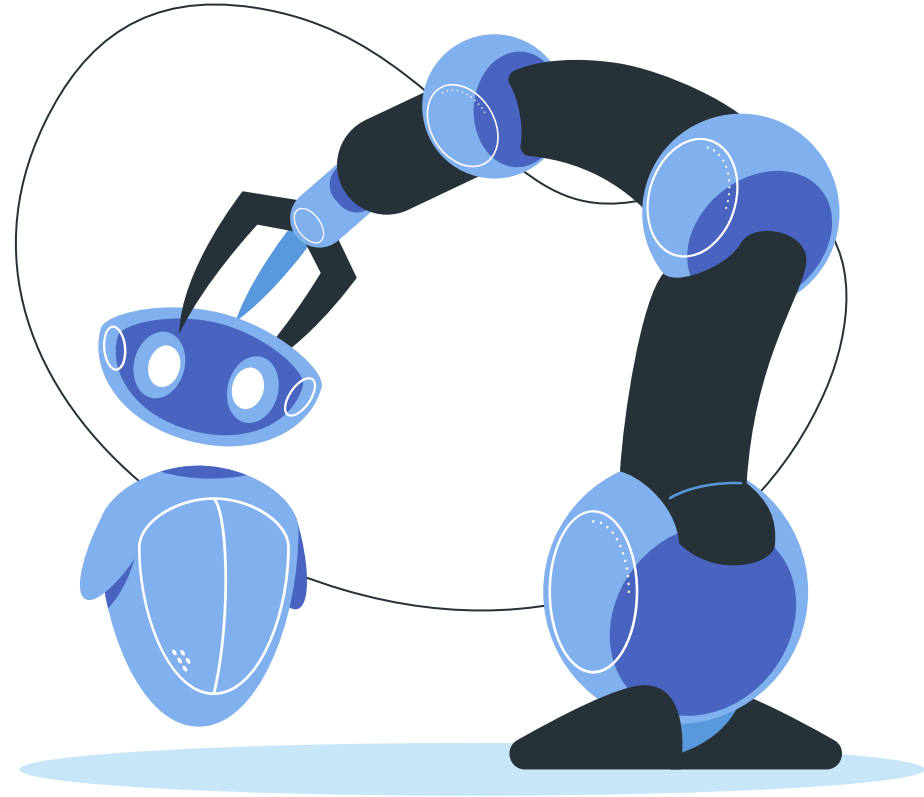


Final Presentation

Friday 9–11



Agenda

1 Design Ideas

2 Strategy

3 Implementation requirements

4 Information section

Strategy

UR5

Bottle pour and stirring cup (2 pts)

- Pick up bottle and pour into cup
- Bonus point for stirring

Microwave tasks (max 12 pts)

- Pick up plate while opening microwave
- Microwave plate food, press button, put on tray
- Pick up bowl while opening microwave
- Microwave bowl food, press button, put on tray

Mobile Robot

Complete all expected tasks (7 pts)

- Four tray points
- Camera to avoid obstacles (3 pts)

Ramp tasks (max 6 pts)

- Low COM allows ramp bonus points

UR5 Strategy

Computer vision

- Detect specific colored circular objects (cup, bowl, plate)
- Detect microwave handle and gripper relative to handle using blue tape

Bottle pour and stirring cup

- First task since it's outside of microwave operating area

Microwave tasks

- Pick up plate while opening microwave
- Microwave plate food, put on tray
- Pick up bowl while opening microwave
- Microwave bowl food, put on tray



Syncing Coordinate Frames



Computer vision

- Objects get isolated via HSV → edge detection → Hough transform to detect shapes (circle for bowl, plate, cup; rectangle for microwave tape)
- Shapes now exist in camera's native coordinate frame

UR5 arms' separate coordinate frames

- Each arm has its own native, internal 3D coordinate frame; these are different from each other and also different from the camera's coordinate frame

Coordinate transform using AprilTags

- A generalized coordinate frame was obtained using a large AprilTag at multiple locations + orientations as visualized from the camera
- An AprilTag was attached to each arm and pictures were taken from the camera at multiple locations + orientations to link up all 3 coordinate frames
- 3D transforms were stored locally and utilized whenever CV needed to be implemented

UR5 Software



Computer vision

- Objects get isolated via HSV → edge detection → Hough transform to detect colored shapes in camera frame.
- Calibrated a conversion between camera and robot frame by placing and detecting AprilTag at multiple end effector locations.
- Used detected location of objects to move end effector onto exact object location to adjust for shifts before competition.

Manipulation

- Used waypoints with position control to move arm into safe configurations before using CV to precisely grip objects.
- Used force control to exert a desired contact force on microwave handle and button.

Orchestration

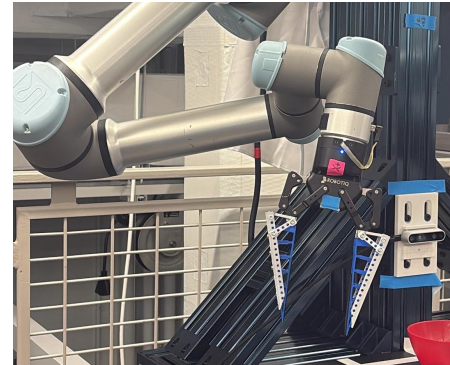
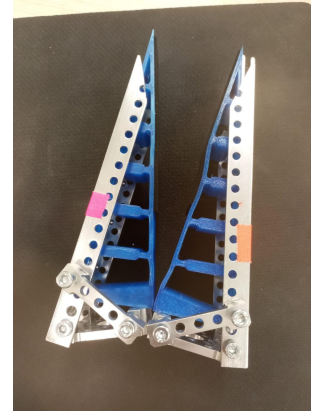
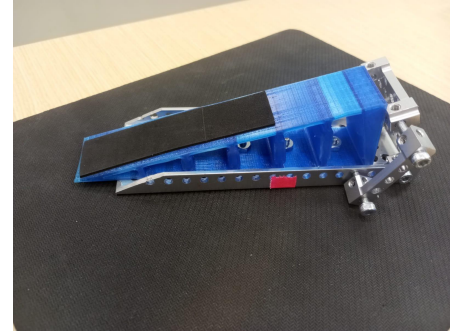
- Created a web interface with a state machine to run tasks simultaneously and enable restart from any task in case of incompleteness.

UR5 Hardware

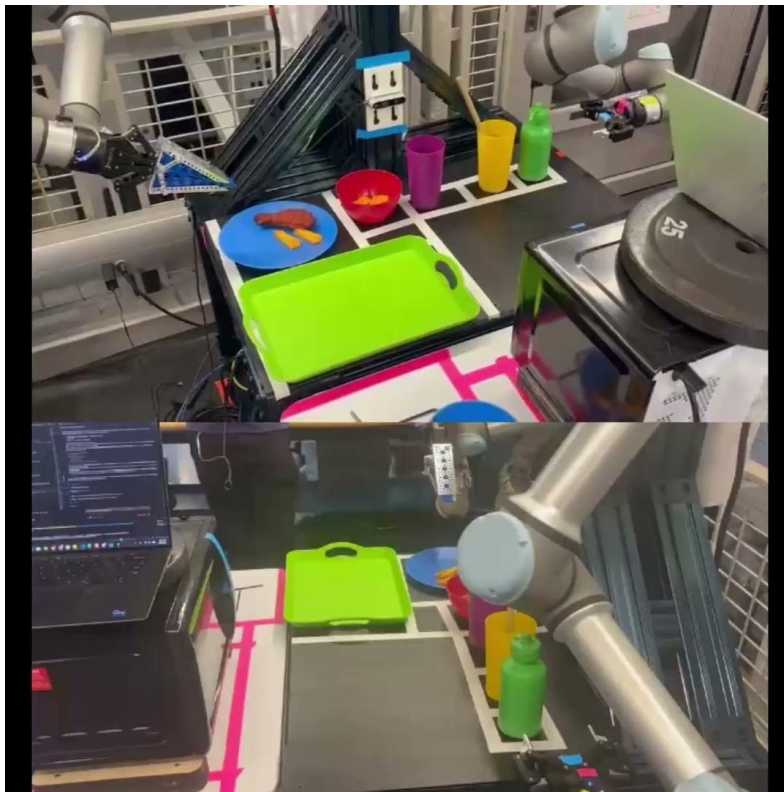


Grippers

- Interior 3D printed TPU 95 grippers with rubber attachments.
 - Fin ray effect for flexibility to conform around objects.
 - Allows spatula manipulation of plate.
- Exterior stainless steel grippers
 - Adjustable joints for rigidity.

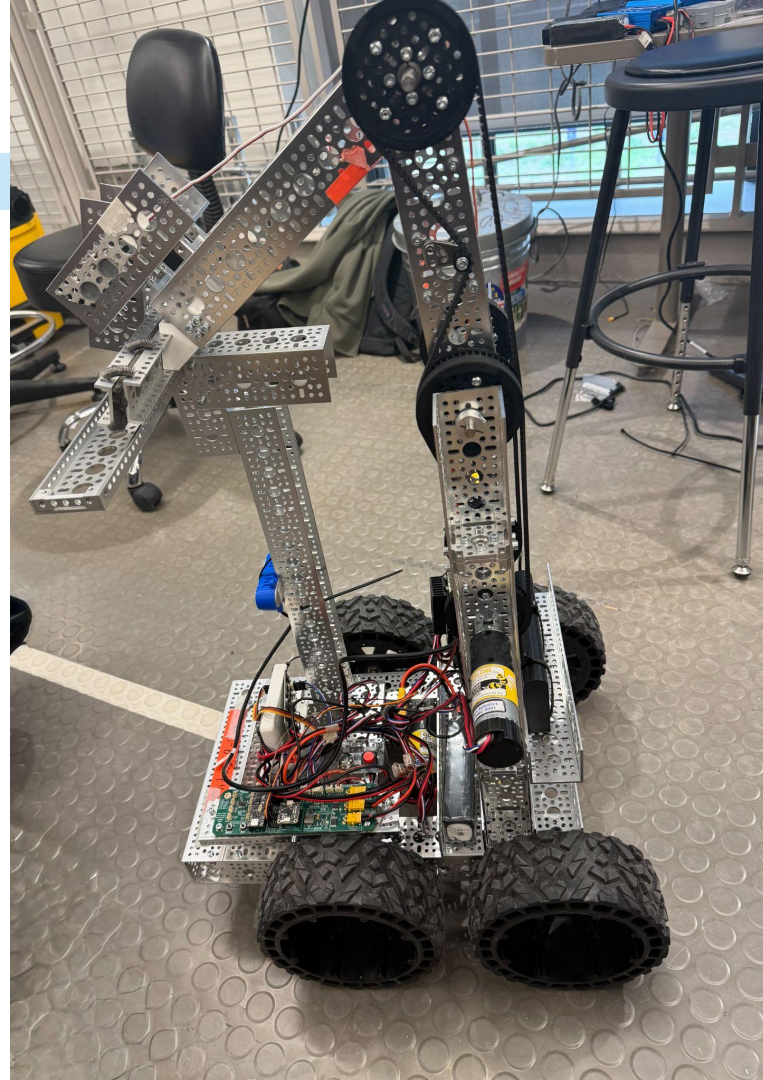


Demo Video



Design

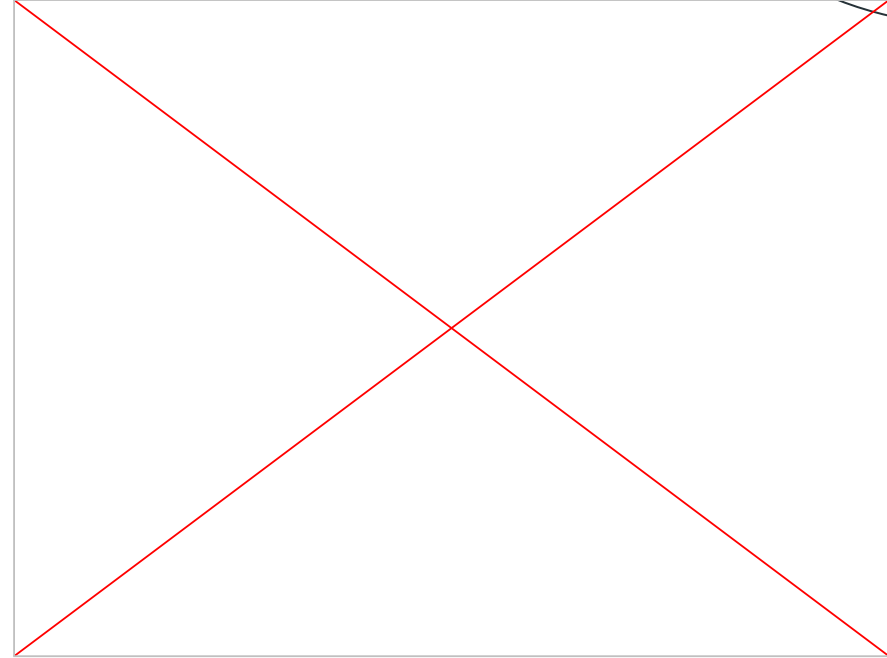
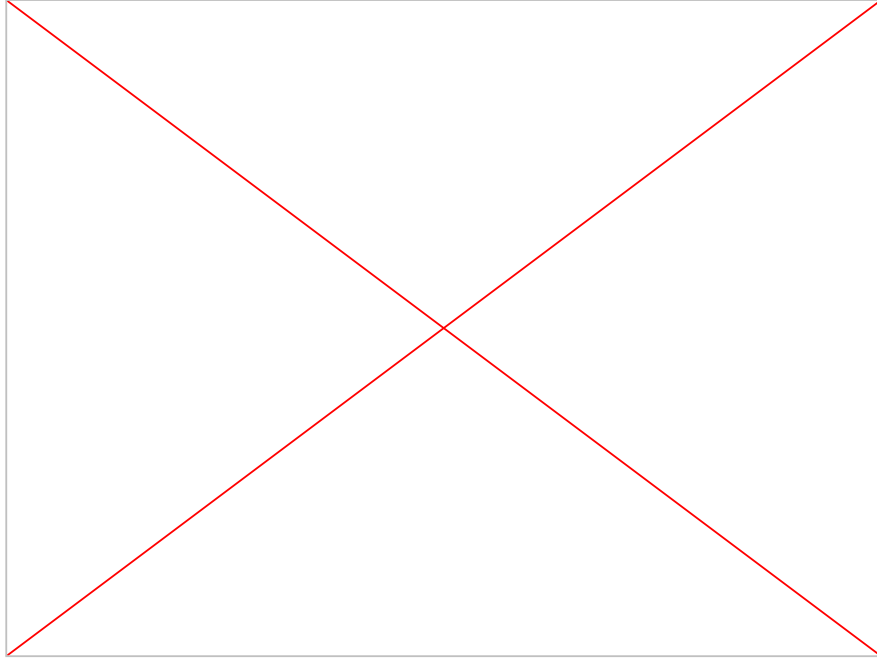
- Focus on simplicity
- 2 DOF arm
 - Belt drive motors close to chassis
- Compact chassis for better odometry
- Use of tray lip as handle
 - Sufficient grip force and friction



Mobile Robot Strategy

- Complete all expected tasks (7 pts)
 - Four tray points
 - Camera to avoid obstacles (3 pts)
- Low COM allows ramp points (6pts)

Demo Videos



Section information

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